

SYSTEM REQUIREMENTS SPECIFICATION (SRS)

CLICK COMPUTER AND ACCESSORIES

ITC327W – Systems Analysis and Design

Version 2.0 – Unit 1 and 2

30 August 2026

Prepared by:

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Nosenga B	223066258	Github, supabase lead

1. Introduction

1.1 Stakeholder and Organisation

Click Computer and Accessories provides computer and electronic support services to clients. Clients may require assistance with cellphones, laptops, printers, televisions and other electronic equipment.

The proposed system is intended to improve how support requests are submitted, monitored, assigned and tracked hence a need for a centralised support platform.

The main stakeholders are clients, support operators, technicians and administrators/managers. The existing process may depend on direct communication with staff, making it difficult to track requests, assign work consistently and maintain visibility of progress.

1.2 Refined Problem Statement

The main problem is the lack of a simple, centralised platform through which clients can select their device, identify the problem or service required, submit a support request and track its progress. The current approach may result in delays, missed requests, unclear technical information, difficulty assigning technicians and limited visibility of request progress.

1.3 Project Aim and Objectives

Aim: To develop a simple client-support platform for Click Computer and Accessories that enables clients to submit, communicate about and track computer and electronic support requests.

Objectives:

1. Provide a simple and user-friendly client application.

2. Allow clients to register and log in, select a supported device, select a relevant problem/service, describe the problem, submit a request and track its status.
3. Allow authorised support operators to monitor incoming requests.
4. Allow support operators to communicate with clients.
5. Allow support requests to be assigned to technicians.
6. Allow administrators to manage users, requests and assignments.
7. Provide notifications for important request events where retained in the final scope.
8. Allow relevant device or asset information to be recorded where required.

1.4 Project Scope

The refined scope prioritises the core support-request process: client registration/login, device selection, problem/service selection, request submission, request status tracking, communication, technician assignment and administrator management. Notifications, asset/device information and supporting evidence remain lower-priority features and may be implemented if time and resources permit.

Included Scope

- Client registration and login.
- Device selection.
- Problem/service selection.

- Support-request submission.
- Request status tracking.
- Client/support communication.
- Technician assignment.
- Administrator controls.
- Device/asset information where required.
- Supporting evidence where retained.

Exclusions

- Unapproved device categories.
- Complex financial/accounting functions.
- Advanced artificial-intelligence diagnosis.
- Automated physical repair of devices.
- Unauthorised access to client devices.
- Features outside agreed stakeholder requirements unless approved through change control.

1.5 Constraints and Assumptions

Technical: Flutter/Dart frontend, C# and ASP.NET Core 6.0 backend, and Supabase persistent storage.

Time: four-week development cycle.

Budget: R0/free-tool approach as stated in the project information. Users are assumed to have suitable internet access, provide accurate information, and use the system according to their roles.

1.6 Deliverables

Project Deliverables

1. Flutter client application.
2. ASP.NET Core backend/API.
3. Supabase database.
4. Client support-request functionality.
5. Request tracking and communication functionality.
6. Technician assignment functionality.
7. Administrator management functionality.

1.7 Definitions

UAT - User Acceptance Testing.

FR – Functional Requirement.

NFR – Non-functional Requirement.

SR – System Requirement.

IR – Integration Requirement.

SRS – System Requirements Specification.

Client – a person who uses the platform to request technical support.

Asset – a device or electronic equipment recorded for support or repair purposes.

2. Stakeholders, Users and Current Process

2.1 Stakeholders and Users

Stakeholder/User	Role
Clients	Submit and track support requests.
Support Operators	Monitor incoming requests, communicate with clients and manage active

	requests.
Technicians	Handle technical problems assigned to them.
Administrators/Managers	Manage users, requests, assignments and service operations.

2.2 Current Process

The existing process may require a client to communicate directly with staff, explain the device and problem manually, wait for the request to be handled and receive feedback through staff communication.

2.3 Current Process Problems

The identified problems include delays in receiving support, difficulty communicating technical problems clearly, requests being missed or forgotten, difficulty assigning requests to appropriate technicians, limited visibility of request progress, increased workload for support staff and poor communication.

2.4 Proposed Process

Client opens application → selects device → selects problem/service → describes and submits request → support operator monitors request → request is assigned to technician where necessary → technician resolves the problem → status is updated → client tracks progress.

3. System Requirements

3.1 Functional Requirements

ID	Requirement	Priority
FR1	The system shall allow clients to register and log in.	Must Have
FR2	The system shall provide an approved device menu for clients.	Must Have
FR3	The system shall display relevant problem/service options after a device is selected.	Must Have
FR4	The system shall allow clients to select a problem, enter a description and submit a support request.	Must Have
FR5	The system shall allow clients to view the current status of a submitted request.	Must Have
FR6	The system shall provide communication between clients and authorised support staff regarding support requests.	Must Have
FR7	The system shall allow authorised users to assign support requests to appropriate technicians.	Must Have
FR8	The system shall allow authorised administrators to manage users, requests and assignments.	Must Have
FR9	The system should send notifications for important request events.	Should Have

FR10	The system should allow relevant device or asset information to be recorded.	Should Have
FR11	The system should allow relevant screenshots, photographs or documents to be attached.	Should Have

3.2 Non-functional Requirements

ID	Requirement	Priority
NFR1	The system shall protect client, technician and administrator information from unauthorised access.	Must Have
NFR2	The system shall provide clear confirmation, status and error messages.	Must Have
NFR3	The system shall maintain accurate and consistent support-request information.	Must Have
NFR4	The client application shall use clear navigation and require minimal steps for submitting a request.	Must Have
NFR5	API responses shall return a result within 2 seconds under normal load.	Must Have

3.3 System/Technical Requirements

ID	Requirement	Priority
SR1	The frontend shall be developed using Flutter and Dart.	Must Have

SR2	The backend shall be developed using ASP.NET Core 6.0 and C#.	Must Have
SR3	The system shall use Supabase for persistent data storage.	Must Have

3.4 Integration Requirements

ID	Requirement	Priority
IR1	Flutter shall communicate with the ASP.NET Core backend through APIs.	Must Have
IR2	ASP.NET Core shall communicate with Supabase to store and retrieve relevant data.	Must Have
IR3	Client support-request information shall pass through the backend and be stored in the database.	Must Have
IR4	Authorised request-status changes shall be reflected in the client application.	Must Have
IR5	Communication and notification events shall be connected to relevant requests and users where implemented.	Should Have

3.5 Acceptance Criteria

Requirement	Acceptance Criteria
FR1	A client can successfully register and log in.
FR2	The device menu displays approved device categories.

FR3	Selecting a device displays relevant support options.
FR4	A client can enter a problem and submit a request.
FR5	A client can view the current request status.
FR6	A client and authorised support operator can communicate about a request.
FR7	An authorised user can assign a request to a technician.
FR8	An administrator can manage users, requests and assignments.
FR9	Relevant users receive notifications if the feature is implemented.
FR10	Device/asset information can be recorded where required.
FR11	Supporting evidence can be attached where the feature is implemented.

3.6 Requirements Traceability table

Stakeholder Need	Requirements
Clients need to submit support request	FR1. FR2, FR3. FR4
Clients need to know request status	FR5
Clients need t communicate with support staff	FR6
Support staff need to assign technician	FR7
Admins need to manage requests	FR8
Users need secure access	NFR1
Users need simple navigation	NFR4

4. Feasibility Analysis

Feasibility Area	Decision	Main Reason	Change Required
1.Technical	Feasible	Separate Flutter/Dart frontend, ASP.NET Core 6.0 backend, and Supabase	Mitigate integration risks strictly within the restricted development timeline.
2.Economical	Feasible	The project will primarily use free/open-source development tools and available services. This supports the project's R0 budget constraint. A limitation is that some hosted services may have usage restrictions or paid tiers	The team will prioritise free service tiers and replace any paid dependency if it becomes necessary.
3.Operational	Feasible	The proposed system is suitable for Click Computer and Accessories because it centralises support requests, allows support operators to monitor requests and enables clients to track progress.	The main limitation is that staff and clients will need to become familiar with the new workflow. Short user training and clear instructions will therefore be required..
4.Schedule	Feasible	Significate features can be delivered in a realistic timeline using Microsoft Project	We consider using the MS Project schedule to track progress,

		Schedule.	milestones and avoid feature creep.
5.Legal	Feasible	System will handle personal data and support conversations through notifications or emails.	Implementing proper access controls and data-retention policy. Legal review of privacy notice will be recommended.
6.Ethical	Feasible	System is used for legitimate operations. Main risks are misuse of data and lack of transparency.	Defining clear usage policy, role-based access, and monitoring that the system is used fairly.

4.2 Overall Feasibility Decision

The Click Computer and Accessories support-request system is considered **feasible with changes**. The technical, operational, economic, schedule, legal and ethical feasibility assessments indicate that the project can be developed using the proposed Flutter, ASP.NET Core and Supabase technologies. However, changes were required to keep the project realistic within the available time, resources and project scope. These changes include prioritising core support-request functionality, moving simplicity to the non-functional requirements, treating 24/7 active management as an operational requirement, and retaining notifications, asset/device information and supporting evidence as lower-priority Should Have requirements. The refined scope will therefore focus on the core functions required for clients, support staff, technicians and administrators.

5. Project Risks

5.1 Risk Register

Risk ID	Risk Statement	Likelihood	Impact	Score	Level	Mitigation	Contingency	Owner	Status	Category
R01	Clients may struggle to use the interface and submit support requests.	3	3	9	High	Keep the main client journey simple and easy to navigate.	Provide guidance to users and simplify the affected screens if required.	Taaibosch K	Active	Operational
R02	Support requests may be missed or not handled on time.	2	3	6	Medium	Provide active monitoring and notifications where implemented.	Review missed requests and assign them to an available support operator.	Ndlovu N	Active	Operational
R03	A support request may be assigned to the wrong technician.	2	3	6	Medium	Use authorized assignment according to the device and problem.	Reassign the request to the appropriate technician.	Mafunisa T	Active	Operational
R04	Poor communication may result in clients not receiving progress information.	2	3	6	Medium	Provide communication linked directly to support requests.	Contact the client through an alternative approved communication method.	Molefe B	Active	Communication
R05	Clients may provide incomplete technical information about their problem.	3	2	6	Medium	Allow supporting evidence such as screenshots, photographs or documents where implemented.	Contact the client and request the missing information or evidence.	Taaibosch K	Active	Operational
R06	Incorrect request status information may mislead clients.	2	3	6	Medium	Restrict status changes to authorized users.	Correct the status and inform the affected client of the correct progress.	Hlalele D	Active	System
R07	Flutter, ASP.NET Core and Supabase components may fail to communicate correctly.	2	3	6	Medium	Test Flutter, ASP.NET Core and Supabase throughout development.	Identify the failed component and restore communication before continuing development.	Mafunisa T	Active	Technical

R08	Limited development time may result in some requirements not being completed.	3	3	9	High	Prioritize Must Have requirements before lower-priority features.	If a Must-Have feature is not ready by the end of Week 3, the team will inform the Project Manager immediately to re-prioritize the remaining features.	Lesako M	Active	Schedule
R09	The project budget may not support commercial services.	2	2	4	Low	Use free tools and services where possible.	Remove or replace features that require paid services.	Lesako M	Monitoring	Economical
R10	Continuous 24/7 support monitoring may not be maintained.	2	2	4	Low	Designate an active support operator if 24/7 monitoring remains required.	Adjust monitoring responsibilities or operating hours if continuous coverage cannot be maintained.	Nosenga B	Monitoring	Operational

6. Requirements Change Log and Refined Scope

6.1 Requirements Change Log

Area	Decision	Reason
Simple client application	Moved to non-functional usability requirements.	Simplicity describes quality rather than a business function.
Laptop repair examples	Removed as a standalone requirement.	Too specific; covered by problem/service selection.
24/7 active management	Moved to operational requirements.	Continuous staffing is an organisational responsibility.
Notifications	Retained as Should Have.	Useful but lower priority than core request processing.

Asset/device information	Retained as Should Have.	Relevant but lower priority.
Supporting evidence	Retained as Should Have.	Useful for technical diagnosis but can be deferred.

6.2 Refined Project Scope

The final scope prioritises client registration/login, device selection, problem/service selection, request submission, request tracking, communication, technician assignment and administrator management. The system will use Flutter/Dart, ASP.NET Core 6.0/C# and Supabase. Notifications, asset/device information and supporting evidence will be implemented only if time and resources allow.

7. System Architecture and Technology

7.1 System Architecture

The proposed architecture consists of the client application, backend/API and database.

Client → Flutter/Dart Application → ASP.NET Core 6.0 Backend/API → Supabase Database → Support Operator / Technician / Administrator.

7.2 Main Components

Client application: registration/login, device selection, problem/service selection, request submission, request status and communication.

Backend: authentication, request processing, assignment, status management, user management, communication and notifications where implemented.

Database: client information, device information, support requests, request status, technician information, assignments, supporting evidence and communication records.

7.3 Technology Stack

Layer	Technology
Frontend	Flutter (Dart)
Backend	ASP.NET Core 6.0 (C#)
Database	Supabase

7.4 Simple Client User Journey

Step 1 – Open App

Step 2 – Select Device

Step 3 – Select Problem/Repair

Step 4 – Submit Request

Step 5 – Get Help / Track Status

8.Stakeholder Register

NAMES	TYPE	ROLES
1. Prince Morafo	Main Stakeholder - External	Project Stakeholder
2. Lesako M	Internal	Project Manager
3. Hlalele D	Internal	Requirements Analyst
4. Tladi K	Internal	Stakeholder Analyst
5. Taibosch K	Internal	Business/Problem Analyst
6. Molefe B	Internal	Feasibility Analyst
7. Ndlovu N	Internal	Risk Manager
8. Mafunisa T	Internal	Solution/Technical Architect
9. Lesenycho LJ	Internal	Project Planner
10. Winkel K	Internal	Documentation and Quality Controller
11. Nosenga B	Internal	GitHub and Presentation Lead

8.2 STAKEHOLDER EVIDENCE

Stakeholder engagement was conducted during Phase 1 to confirm the current process, stakeholder needs, requirements and proposed scope. The stakeholder provided feedback on the proposed system, and the feedback was used to refine the requirements and project scope.

Supporting evidence of stakeholder engagement, requirements confirmation and feedback has been retained by the project group and will be made available for verification if required. Due to confidentiality and submission requirements, the original evidence is not included in the submitted SRS.

9. Revision History

Version	Date	Changes
1.0	AUG 25	Initial SRS document
2.0	AUG 30	Added Unit 2 feasibility analysis, risk register, refined requirements, change log, refined scope and stakeholder register

10. Approvals

Role	Name	Date
Stakeholder	Morafo Prince	29 August 2026
Project Lead	Lesako Mpho	30 August 2026

11. Appendix – Stakeholder Evidence

A. Stakeholder Engagement Summary

Date	Type	Purpose	Attendees
24 August 2026	Site Visit	Initial stakeholder identification	Mafunisa T, Tladi K
25 August 2026	Site Visit	Stakeholder interview & problem clarification	Mafunisa T, Tladi K
26 August 2026	Online Meeting	Requirements confirmation & sign-off	All 10 members

B. Key Interview Findings

The stakeholder (Prince Morafo) confirmed the following during 25 August 2026 interview:

1. **Current Process Issues:**

- Clients contact the business directly via phone or in-person
- No structured system for tracking requests
- Difficult to assign work to appropriate technicians
- Clients have no visibility into request progress

2. **Stakeholder Requirements Confirmed:**

- Simple client application (REQ – 01)
- Client registration/login (REQ – 02)
- Device menu (REQ – 03)
- Problem/service selection (REQ – 04)
- Simple request submission (REQ – 06)
- Request status tracking (REQ – 07)
- 24/7 active management (REQ – 08)
- Technician assignment (REQ – 10)
- Administrator controls (REQ – 11)

3. **Scope Confirmation:**

- In scope: Core request management functions

- Out-of-scope: Physical repairs, payment processing
- Notifications, asset information, and supporting evidence retained as ‘Should Have’

C. Evidence files

The following evidence files are available in the GitHub repository under `docs/stakeholder/evidence/` :

File	Description
WhatsApp Image 2026-09-03 at 18.38.44	It's a screenshot of when we asked the stakeholder to give us a logo we can use for the project
WhatsApp Image 2026-09-03 at 18.39.34	It's a screenshot of when we asked
Click_Computer_and_Accessories_Stakeholder_Requirements_Signoff (1)	A file that contains the requirements listed and signed off by the stakeholder

D. Change Confirmation

The following changes were confirmed with the stakeholder:

Change	Stakeholder Response
Notifications as Should Have (not Must Have)	Agreed – core functions are priority
24/7 monitoring as operational responsibility	Agreed – business will ensure staffing
Supporting evidence as Should Have	Agreed – can be added later
Laptop examples removed as separate requirements	Agreed – covered by problem/service selection

E. GitHub Repository Evidence

The group's GitHub repository contains all project documentation and shows evidence of team activity.

Repository URL: <https://github.com/tmafunisa24-sudo/ITC327W-Project>

Figure E.1 : GitHub Repository Overview

Located at (./docs/evidence/screenshots/github_repo_overview.png)

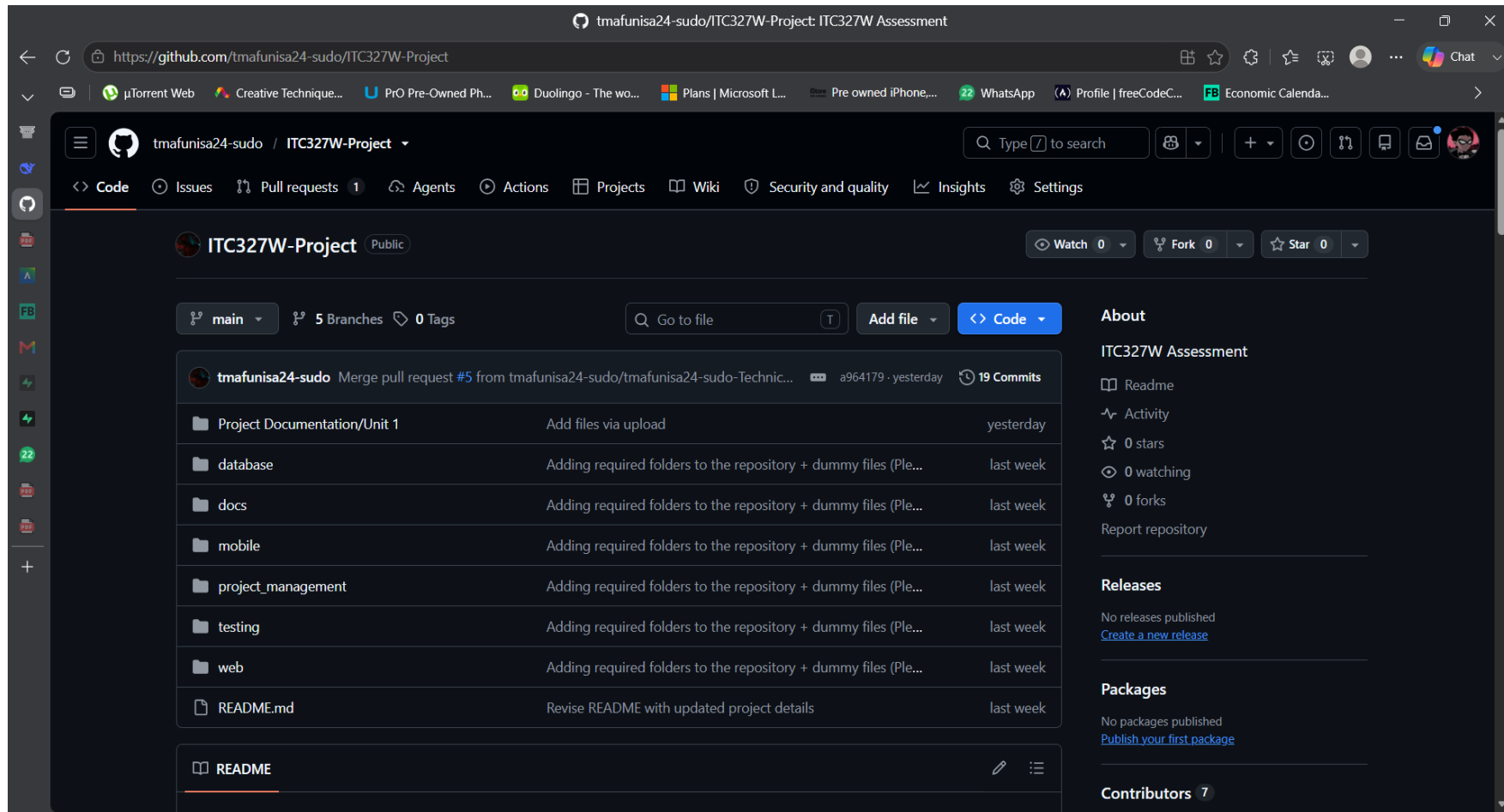


Figure E.2 : Commit History

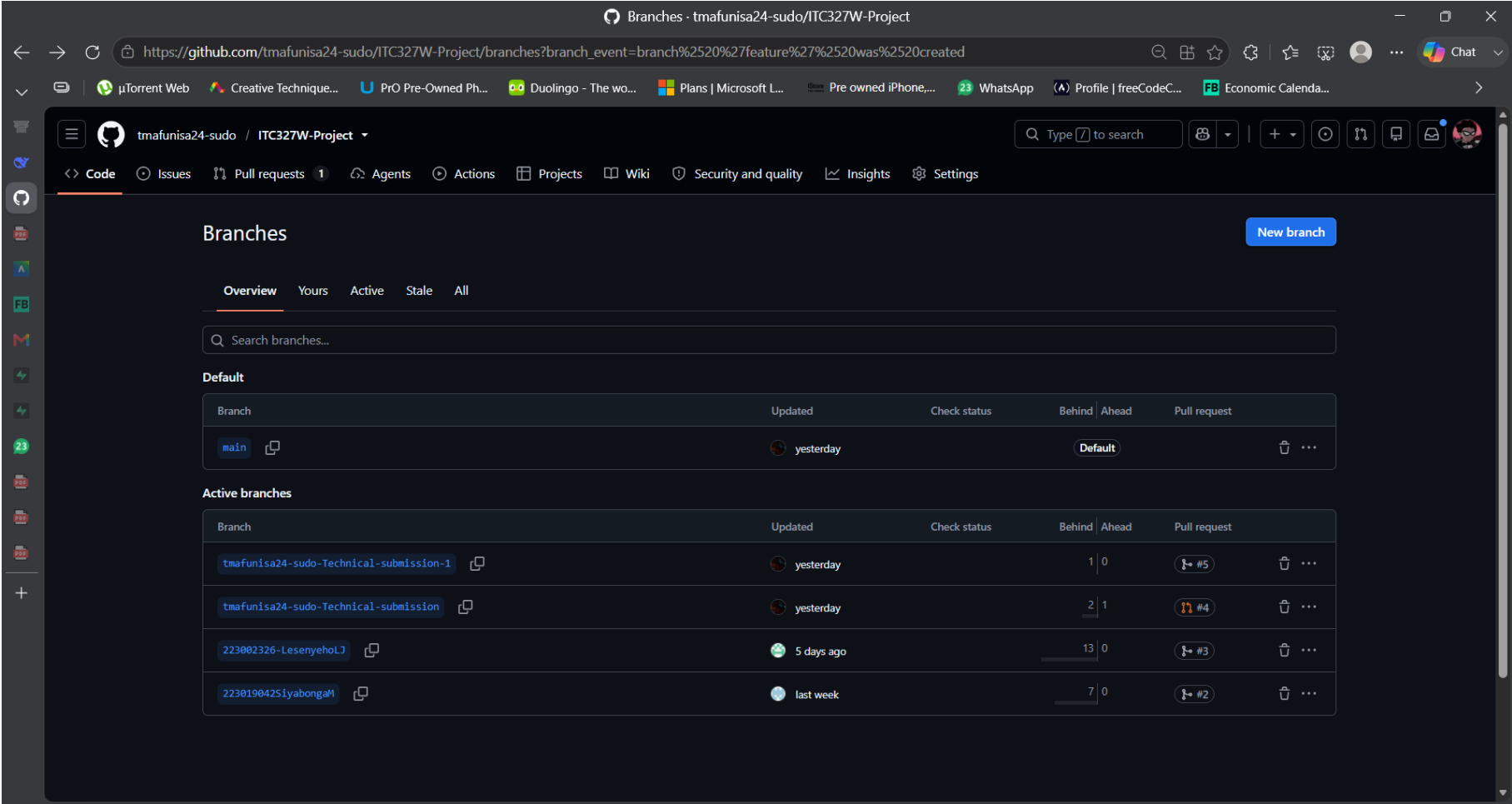
Located at (./docs/evidence/screenshots/github_commit_history.png)

The screenshot displays the GitHub commit history for the repository `tmafunisa24-sudo/ITC327W-Project`. The interface is in dark mode. The top navigation bar includes the repository name and a search bar. The left sidebar shows the repository's navigation menu with options like Code, Issues, Pull requests, Agents, Actions, Projects, Wiki, Security and quality, Insights, and Settings. The main content area is titled 'Commits' and shows a list of commits grouped by date. The commits are as follows:

- Commits on Sep 3, 2026**
 - Merge pull request #5 from tmafunisa24-sudo/tmafunisa24-sudo-Technical-submission-1** (Verified, a964179) by tmafunisa24-sudo, authored yesterday.
 - Add files via upload** (Verified, 9472364) by tmafunisa24-sudo, authored yesterday.
 - Merge pull request #2 from tmafunisa24-sudo/223019042SiyabongaM** (Verified, 8f0c3ff) by scroogerzy, authored yesterday.
- Commits on Aug 31, 2026**
 - Merge pull request #3 from tmafunisa24-sudo/223002326-LesenyehoLJ** (Verified, 13ffb9) by mphoLesako, authored 4 days ago.
 - Merge branch '223002326-LesenyehoLJ' of https://github.com/tmafunisa24-sudo/ITC327W-Project into 223002326-LesenyehoLJ** (Verified, 5bc1daa) by LitebohoJL, committed 4 days ago.
 - Refined Microsoft Project** (Verified, 16f262c) by LitebohoJL, committed 4 days ago.
- Commits on Aug 27, 2026**
 - Revise README with updated project details** (Verified, 63f3c71) by Nosenga, authored last week.
- Commits on Aug 26, 2026**
 - Added feasibility document** (Verified, e612ea7) by molefesiyabonga34-droid, committed last week.

Figure E.3: Branch Structure

Located at (./docs/evidence/screenshots/github_branches.png)



Repository Activity Summary:

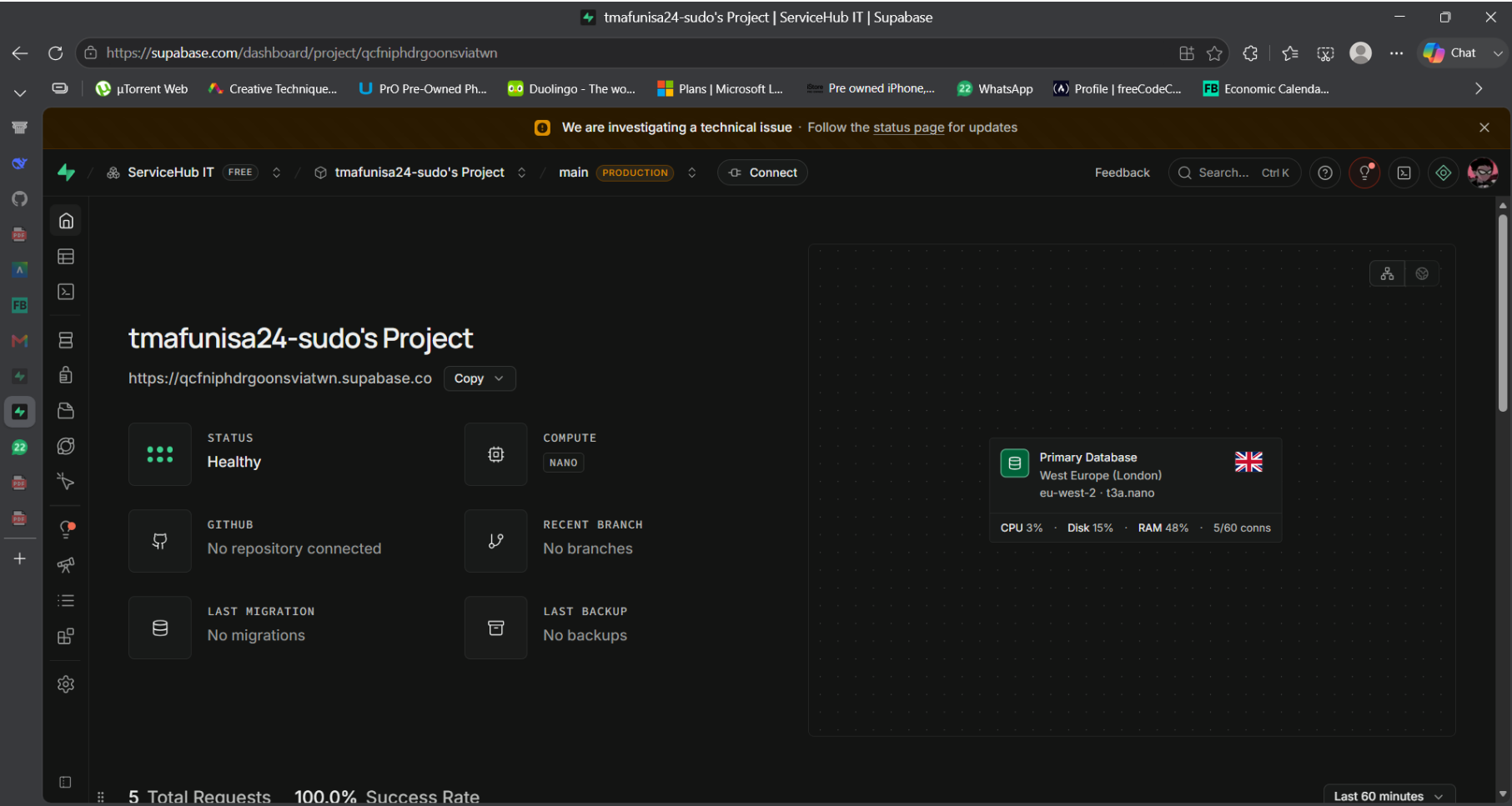
Metric	Status
Repository Created	25 August 2026
Contributors	All 10 members
Branches	`main`, `tmafunisa24-sudo-Technical-submission-1`, `223002326-LesenyehoLJ`, `223019042SiyabongaM`
README	Contains project title, description, technologies, team members, and current stage
Documentation	Currently and organised in the `docs/` folder

F. Supabase Project Setup Evidence

The group created a Supabase project for the backend database

Figure F.1: Supabase Project Dashboard

Located at (./docs/evidence/screenshots/supabase_project.png)



Supabase Project Details:

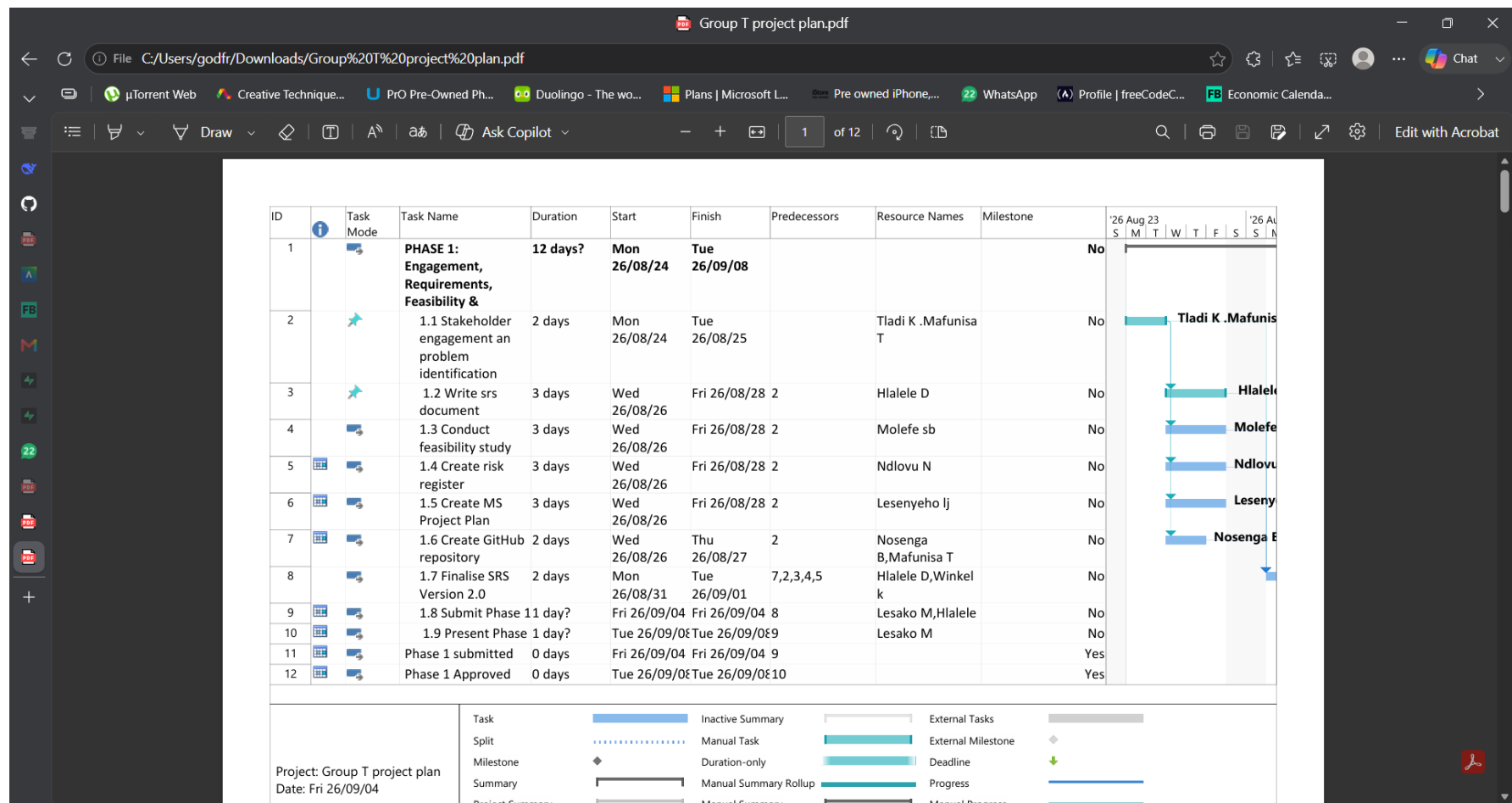
Detail	Information
Project Name	Click Computer and Accessories
Database Type	PostgreSQL
Status	Healthy
Authentication	Supabase Authentication
Storage	Available for future use

G. Microsoft Project Schedule Evidence

The group created a project schedule using Microsoft Project.

Figure G.1: MS Project Gantt Chart

Located at (./docs/evidence/screenshots/ms_project_gantt.png)



Milestone Summary:

Milestone	Target Date	Status
Stakeholder Confirmed	25August 2026	Completed
Requirements Draft	25August 2026	Completed
Feasibility and Risk Done	30September 2026	Completed
Full Draft Review	02September 2026	Completed
Final Submission	04 September 2026	On Track